MATH I - Week 6

Fall 2026

NOTES on continuity, differentiability, left- and right-hand derivatives, kink points, cusps, and vertical tangents.

Prepared by G. Neofotistos (in conversations with OpenAI model 6.1 Sol)

Continuity means the function's values join without a break. Differentiability means its graph has a well-defined, finite tangent slope. A function can be continuous yet fail to be differentiable at a sharp point or vertical tangent.

1. Continuity and differentiability

A function f is **continuous at** $x = a$ if

$$\boxed{\lim_{x \to a} f(x) = f(a).}$$

This requires that $f(a)$ exists, the limit exists, and they agree.

A function is **differentiable at** $x = a$ if the following limit exists as a finite real number:

$$\boxed{f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.}$$

The quotient is the slope of a **secant line** joining two points on the graph. As the second point approaches the first, the secant slope approaches the **tangent slope**, provided the derivative exists.

The fundamental relationship is

$$\boxed{\text{Differentiable at } a \implies \text{Continuous at } a.}$$

The reverse implication is false. For example, $f(x) = |x|$ is continuous at 0, but its sharp corner prevents differentiability there.

Why does differentiability imply continuity? Because

$$f(a+h) - f(a) = h\left(\tfrac{f(a+h)-f(a)}{h}\right).$$

If the quotient approaches a finite number, the product approaches 0. Thus $f(a+h) \to f(a)$.

2. Left- and right-hand derivatives

We can approach a point from either side:

$$\boxed{f'_-(a) = \lim_{h \to 0^-} \frac{f(a+h) - f(a)}{h}} \qquad \text{(left-hand derivative)}$$

$$\boxed{f'_+(a) = \lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h}} \qquad \text{(right-hand derivative).}$$

At an interior point of the domain,

$$\boxed{f'(a) \text{ exists} \iff f'_-(a) = f'_+(a) \text{ and both are finite.}}$$

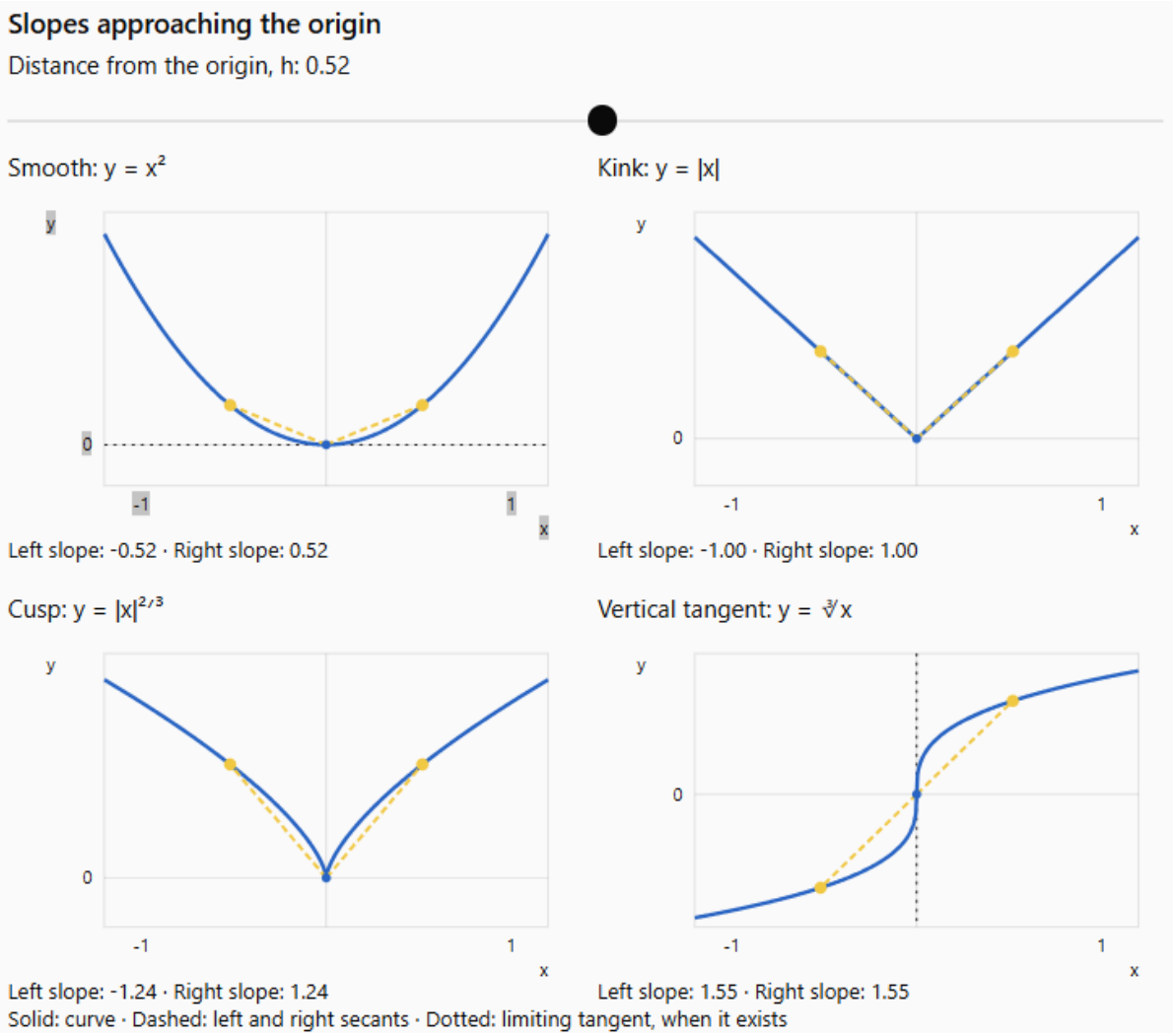
For example, if $f(x) = x^2$, then at $a = 0$,

$$\frac{f(h)-f(0)}{h} = \frac{h^2}{h} = h.$$

From either side, $h \to 0$, so $f'(0) = 0$: the tangent is horizontal.

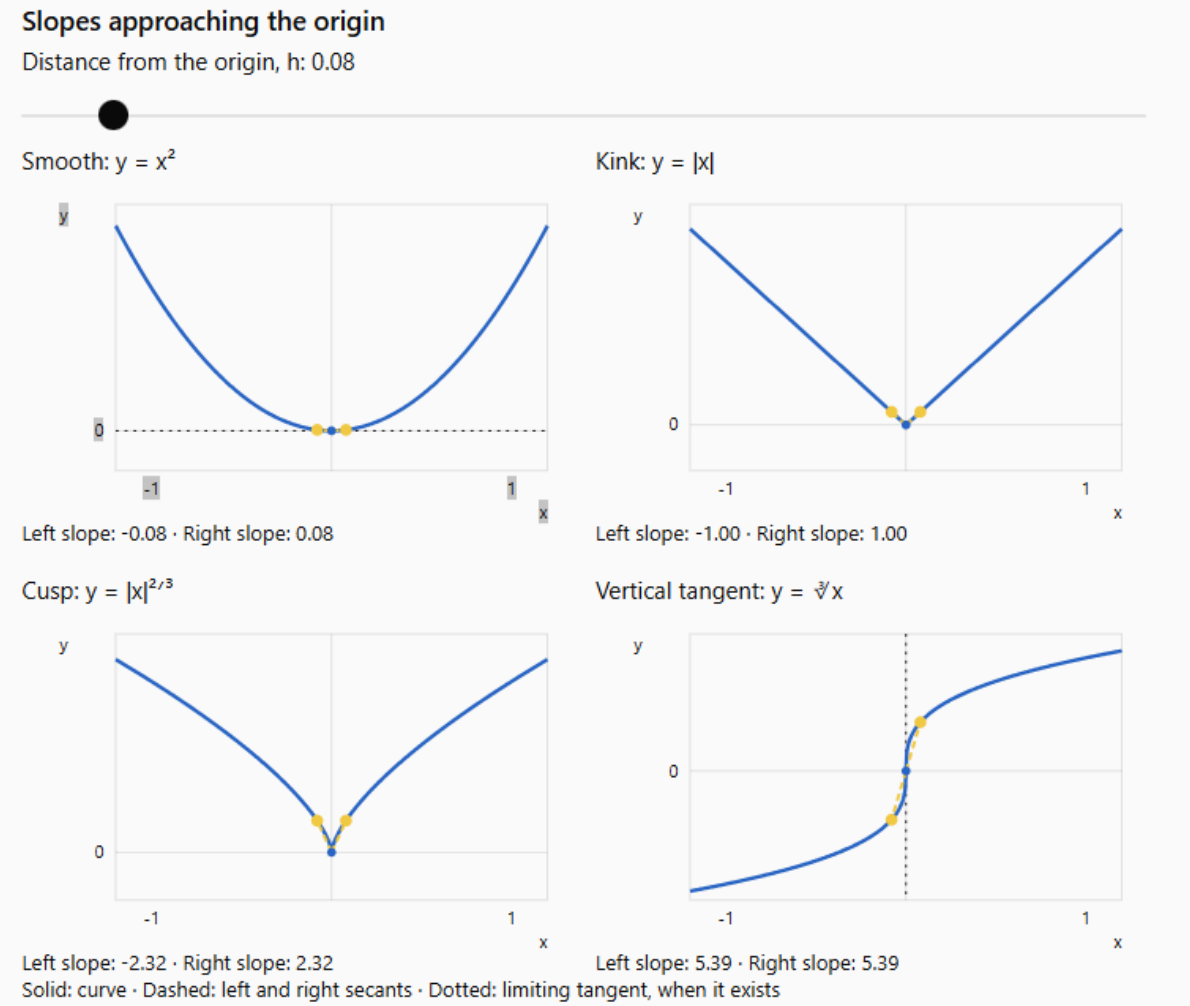
3. See the four main cases

Move the slider toward zero. The dashed secants approach the origin from the left and right. Compare their displayed slopes.



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4. Kink points: two different finite slopes

Consider

$$f(x) = |x| = \begin{cases} -x, & x < 0, \\ x, & x \geq 0. \end{cases}$$

At 0,

$$f'_-(0) = -1, \qquad f'_+(0) = 1.$$

Both slopes are finite, but they disagree. Therefore, f **is continuous but not differentiable at** 0. The graph has a **kink**, also called **a corner**.

For an engineering interpretation, imagine a position graph whose slope changes abruptly from one finite velocity to another.

5. Cusps: opposite infinite slopes

Consider

$$f(x) = |x|^{2/3}.$$

At 0, the difference quotient is

$$\frac{f(h)-f(0)}{h} = \frac{|h|^{2/3}}{h}.$$

Consequently,

$\lim_{h\to 0^-} \frac{|h|^{2/3}}{h} = -\infty, \qquad \lim_{h\to 0^+} \frac{|h|^{2/3}}{h} = +\infty.$

The graph becomes infinitely steep on both sides, with opposite signs. It forms a **cusp**.

The function is continuous at 0, but **no finite derivative exists there**.

6. Vertical tangents: the same infinite slope

Consider

$f(x) = \sqrt[3]{x}.$

At 0,

$\frac{f(h)-f(0)}{h} = \frac{\sqrt[3]{h}}{h} = \frac{1}{|h|^{2/3}}.$

From both sides,

$\lim_{h\to 0^-} \frac{1}{|h|^{2/3}} = \lim_{h\to 0^+} \frac{1}{|h|^{2/3}} = +\infty.$

The graph has the **vertical tangent** $x = 0$. Unlike the cusp, it passes through the point with the same direction of infinite slope.

Although we sometimes informally write "$f'(0) = +\infty$," **the derivative does not exist as a finite real number**.

Example at $x = 0$	Left slope limit	Right slope limit	Continuous?	Differentiable?
x^2	0	0	Yes	Yes
$\lvert x\rvert$: kink	-1	1	Yes	No
$\lvert x\rvert^{2/3}$: cusp	$-\infty$	$+\infty$	Yes	No
$\sqrt[3]{x}$: vertical tangent	$+\infty$	$+\infty$	Yes	No

A useful procedure: first check continuity. If continuity fails, differentiability fails. If continuity holds, compare the left- and right-hand difference quotients: differentiability requires the same **finite** limit.

These examples illustrate common failures of differentiability; other functions can fail because their slopes oscillate without approaching any limit.

These Notes have been excerpted -for the Math I (Fall 2026) course delivered at the Tashkent University of Technology- from Prof. G. Neofotistos book (in preparartion) titled "Calculus for the 21st-century Engineer - Mathematics, Computation, and Artificial Intelligence (AI)". This work is licensed under a Creative Commons "Attribution-NonCommercial-NoDerivatives 4.0 International" license.

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